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Crop protection :: IPM :: Soybean

Monitoring for Pest & Disease

- Undertake community level monitoring to know change in distribution and abundance of pest.
- Organise regular pest monitoring and assess bio control potential at every 15 days interval.
- Record pest and disease incidence at every 5 to 10 km distance at 10 spots/ha and 5 plants/spot selected randomly.
- Install sex pheromone trap for early deduction of *S. litura* @ 10 traps/ha for mass trapping.
- Light trap with 125 watts mercury vapour lamp can be operated for two hours in the evening to observe phototropic insect pests.

Pre-sowing stage

- Deep ploughing in summer.
- Pre-monsoon sowing MUST BE avoided.
- Fluchloralin (1.0 to 1.5 kg a.i./ha) or trifluralin (0.96 to 1.2 kg a.i./ha) should be incorporated into the soil for controlling broad leaved weeds.
- Narrow row spacing (20 cm) + higher seed rate (125 kg/ha) + Oxadiazon 1.0 kg Pre emergence (PE).
- Solarization - Practice soil solarization by giving a very light irrigation in summer morning and the covering the field by polythene, by this the irrigation water evaporates and kills the weed seeds by steaming affect.

Sowing stage

- Use insect/disease tolerant varieties.
- Use optimum seed rate (70-100 kg/ha).
- Avoid pre monsoon sowing.
- Do not cultivate when the foliage is wet.
- Avoid excess nitrogenous fertilizers.
- Application of potash fertilizers should be ensured.
- Intercrop soybean either with (early maturing) Pigeon Pea variety or maize or sorghum in the sequence of 4:2 should be practised.
- Plant seeds relatively free of the pathogen.
- Rotate soybean with crops not susceptible to the pathogen.
- N, P, K, and S should be applied @ 20:60- 80: 20:20 kg per ha.
- Crop should be maintained weed-free initially for 30-45 days by resorting to two hand hoeing /weedings.
- Seed treatment with thiram 75% DS @ 3 g/kg seed should be done. This should be followed by seed treatment with *Rhizobium japonicum* and Phosphate Solubilizing Bacteria (PSB) @ 5 + 5 g/kg seed.
- Pre-emergence application of alachlor (2.5 kg a.i./ha) or metalachlor (1.0 kg a. i./ha) for grassy weeds and pendimethalin (0.75 to 1.0 kg a.i./ha) or metribuzin (0.35 to 0.525 kg a.i./ha) for controlling both grassy and broad leaved weeds.
- Apply phorate 10 % CG @ 10 kg/ha at the time of sowing
- Rogue out *Sclerotium* affected seedlings.

Vegetative stage

- Collect and destroy girdle beetle infested plant parts, egg masses and gregariously caterpillar feeding larvae of hairy caterpillar and tobacco caterpillar.
- Conserve biological control such as spiders, lizards, coccineid, beetles, technid fly, dragon fly, damsel fly, praying mantis.
- Apply oxydemeton methyl* 25% (@ 500- 1000 ml/ ha or dimethoate* 30% EC @ 750 -1000 ml/ha two weeks after germination in case phorate has not been applied at the time of sowing.
- Apply mancozeb 75% WP @ 1500 - 2000 g/11a or hexaconazole 5% EC or propiconazole or triadimefon 80% EC @ 0.1% on the appearance of rust infection.

Flowering stage

- Apply triazophos 40% EC @ 625 ml/ha or quinalphos 25% EC @ 1000 ml/ha for controlling defoliators, tobacco caterpillars, stem fly and girdle beetle.
- Apply mancozeb 75% WP (@ 1500-2000 g/ha or hexaconazole 5% EC or propiconazole 25% EC or triadimefon 25% WP @0.1% on the appearance rust infection.

Fruiting stage

- Poison baiting with 2% zinc phosphide at podding and green seed stage preceded by one day pre- baiting or application of bromadiolone 0.005% ready to use at green seed stage for the control of rodents.

Storage

- Seeds should be harvested at field maturity.
- Delayed harvesting should be avoided.
- Seeds should be dried to safe moisture levels soon after harvest (8-10%).
- The dried seeds should be stored in hermetically sealed containers.
- Protect the sealed containers from water, cross contamination and rodents.
- Ensure proper ventilation around seed.
- Completely cover plant residue by clean ploughing the field soon after harvest.

